



Design and Evaluation of AI-Driven Intrusion Detection System for UAV Swarm Networks

Arshia Naveed TC-22020 Anees-Ur-Rehman TC-22036 Malaika Ghazanfar TC-22046 Shaista Hassan TC-22049

Supervisor: Dr. Hira Mariam | Industrial Advisor: Hafiz Muhammad Attaullah, Senior Cybersecurity Consultant, AI EYS

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INTRODUCTION

Drones are becoming increasingly popular in applications such as security, surveillance, and rescue operations. As drone swarms grow more autonomous, their communication networks become high-value targets for adversarial cyber threats.

Current IDS solutions are NOT designed for the dynamic, distributed nature of swarm-based UAV communication.

- High False Alarm Rates in existing systems disrupt operations.
- Slow Response Times are incompatible with real-time swarm requirements
- Poor Adaptability unable to keep pace with evolving and swarm-specific attack patterns
- Hardware Dependency most solutions are tied to specific platforms, limiting cross-UAV deployment

PROBLEM STATEMENT

Swarm based Unmanned Aerial Vehicle (UAV) communication networks are attractive targets for hackers, who may manipulate flight routes, issue incorrect commands, or disrupt communication channels, potentially compromising safety, mission objectives, and sensitive data.

Targeted attack vectors in UAV swarm networks:

Denial of Service

False Data Injection

Evil Twin Attack

Replay Attack

AIM & OBJECTIVES

This research aims to create and test an advanced AI powered IDS designed specifically for UAV swarm networks. The goal is to enable the system to detect and classify cyberattacks in real time, improving the overall security and reliability of UAV swarm operations.

O1

To design a lightweight AI driven IDS for UAV swarm networks.

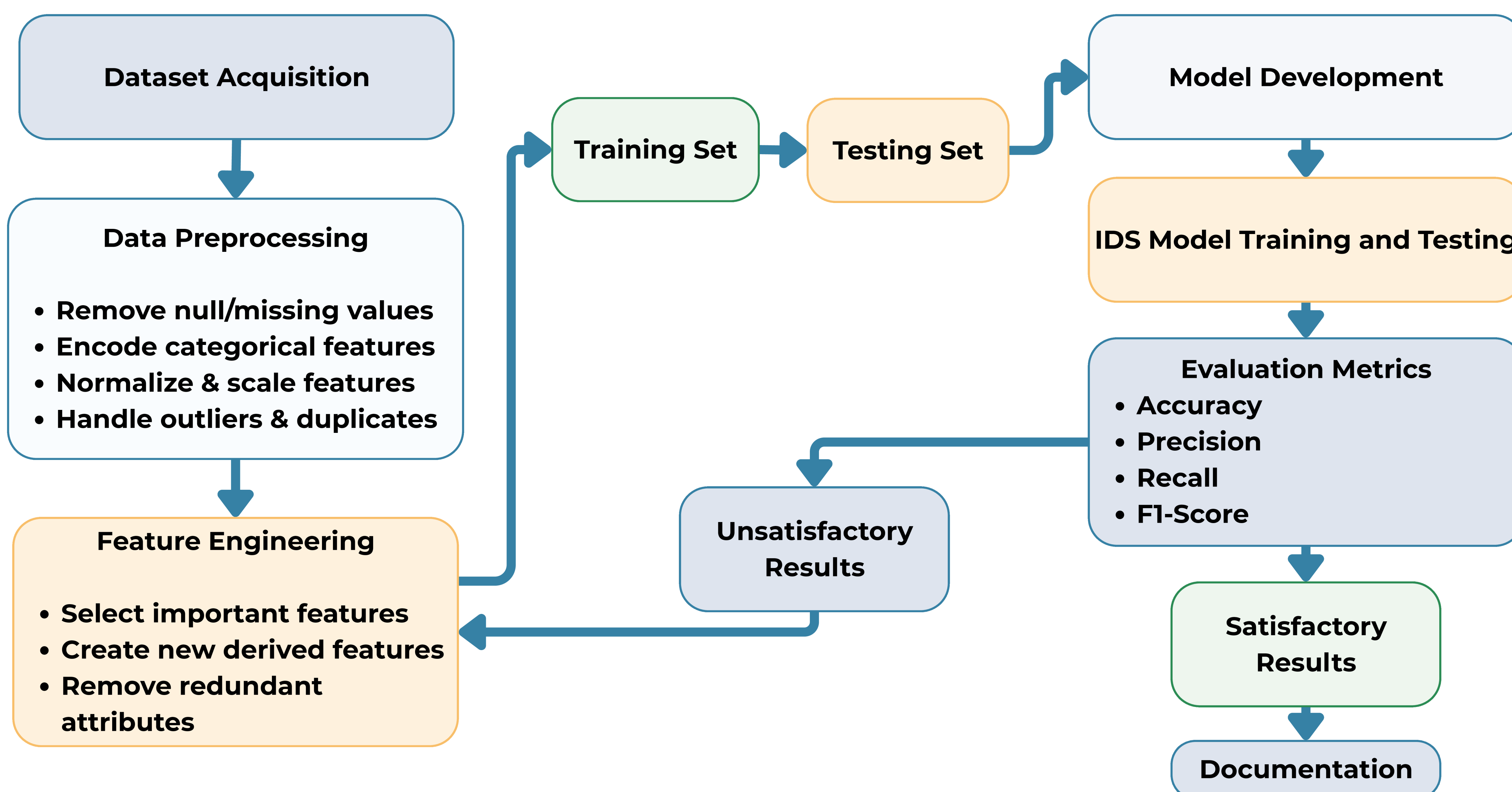
O2

To detect cyberattacks (such as Denial of Service, False Data Injection, Evil twin, Replay attack).

O3

To evaluate the IDS performance using standard detection metrics.

PROPOSED METHODOLOGY



TOOLS & TECHNOLOGIES

Python
Core Language

Jupyter / Colab
Development
Environment

Scikit-learn
ML Classifiers

TensorFlow / Keras
Deep Learning

Pandas / NumPy
Data Processing

Matplotlib
Visualization

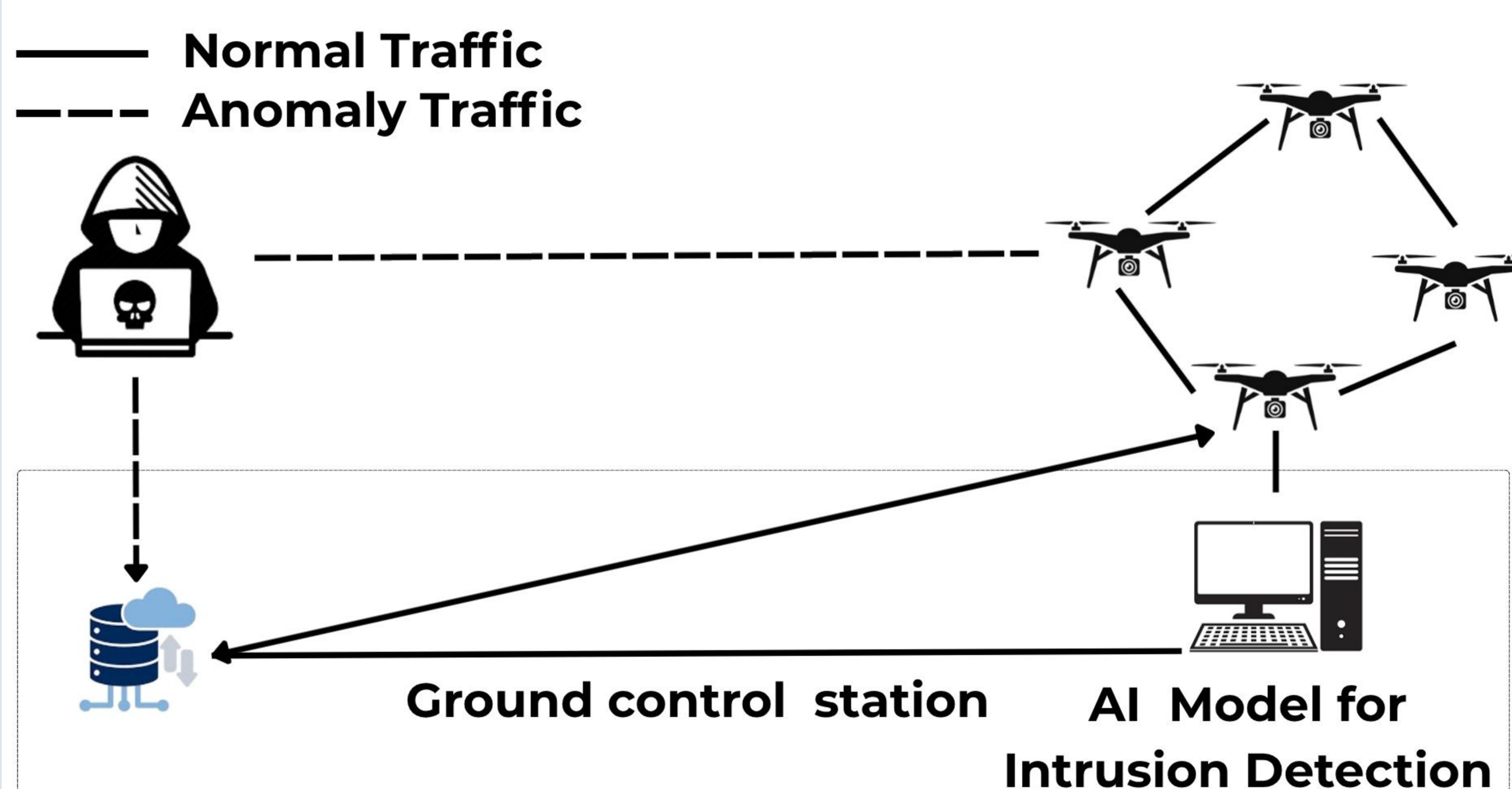
SDG ALIGNMENT

SDG 9
Industry, Innovation &
Infrastructure

SDG 11
Sustainable Cities &
Communities

This project contributes to resilient infrastructure and safer autonomous urban systems by developing robust AI-powered security solutions for emerging UAV technologies.

SYSTEM ARCHITECTURE



DATASET DESCRIPTION

HASSLER, MUGHAL & ISMAIL | IEEE DATAPORT (2023)

UAV Platform DJI Tello EDU | 80g, 802.11n WiFi, UDP Protocol

Total Samples (Rows) 54,783

Total Samples (Columns) 38 Total

Network Topology Static Star GCS = central node, UAV connected via WiFi AP

Attack Layer Layer 2 Attacks DoS, Replay, Evil Twin | IEEE 802.11 frames
Application Layer Attack False Data Injection | UDP, IP 192.168.10.1, Port 8890

Attack Classes Denial of Service, Replay, Evil Twin, False Data Injection